

Air pollution in three megacities of India during the Diwali festival amidst COVID-19 pandemic

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ABSTRACT

The present study characterized the difference in air pollution levels between Diwali celebrations amidst the Covid-19 pandemic (14 November 2020) and the previous year (27 October 2019). The concentration of seven principal air pollutants, namely PM_{2.5}, PM₁₀, NO₂, NH₃, SO₂, CO, and O₃, was substantially higher in 2020 Diwali than in 2019 Diwali. PM_{2.5}, PM₁₀, and CO were always above the permissible limits in all three megacities, with the highest concentrations observed in Delhi. This observation indicates that amidst the pandemic, people burnt more firecrackers than the preceding year. However, the pre-Diwali to Diwali changes in pollutant concentrations suggested that Mumbai and Kolkata had lower pollution levels due to the fireworks. The firework-induced air pollutant load in Delhi was enhanced by a higher margin in the pandemic year. The difference in meteorological conditions between the two consecutive year's Diwali might have led to the enhanced pollutant load in 2020, especially in Delhi. Delhi showed a tendency to retain the pollutants in the ground-level atmosphere for almost one week after the event. Thus, this study implies the need for stringent law enforcement to ameliorate the pollution levels due to such celebrations.

1. Introduction

Diwali, also known as Deepavali, is a festival of lights. In India, people celebrate this auspicious occasion during the post-monsoon months of October or November (Ambade, 2018; Mukherjee et al., 2018). The date of the festival in a particular year varies according to the Hindu lunisolar calendar (Dershowitz and Reingold, 2009). Diwali symbolizes the spiritual victory of light over darkness, good over evil, and wisdom over ignorance (Mathur, 2021), and its celebration involves burning firecrackers and sparklers (Sateesh et al., 2018; Ghei and Sane, 2018). Several festivals in different corners of the world like the New Year celebrations, the Lantern Festival in China, Sky fest in Ireland, and others, include firecrackers (Ambade (2018) and the references therein). However, Diwali deserves a special mention, as hundreds of millions of people participate in burning firecrackers. Across the length and breadth of India, several townships, cities, and metropolises, like Ahmedabad (Ganguly et al., 2019), Bangalore (Gowda et al., 2020), Bhopal (Choudhry et al., 2018), Bhubaneswar (Mandal et al., 2020), Chennai (Prakash et al., 2019), Dehradun (Prabhu et al., 2019), Delhi (Mukherjee et al., 2018; Saxena et al., 2020; Singh and Srivastava, 2020; Patel et al., 2021), Faridabad (Sharma et al., 2018), Guwahati (Garaga and

Kota, 2018), Hyderabad (Chen et al., 2020), Jabalpur (Srivastava et al., 2014), Jaisalmer (Mahecha et al., 2012), Jamshedpur (Ambade, 2018), Jhansi (Chauhan et al., 2014), Kolkata (Sahu, 2019), Lucknow (Barman et al., 2009), Mumbai (Nanda et al., 2018), Nagpur (Kumar et al., 2017), Nashik (Dhanwate, 2017), Sambalpur (Sahu et al., 2020), Udaipur (Chittora and Kapoor, 2015), Varanasi (Kumar et al., 2016), and Vishakhapatnam (Ganguly et al., 2019) recorded a significant rise in air pollutants in the lower troposphere during the Diwali celebrations.

The burning of firecrackers releases an array of harmful chemical compounds (barium nitrate; potassium chlorate, nitrate, and perchlorate; sodium oxalate; strontium nitrate), metals (aluminum, iron oxides, and manganese), metalloids (arsenic), and non-metals (sulfur) into the ambient atmosphere (Kulshrestha et al., 2004; Rajendran et al., 2021). Several flying firecrackers introduce highly toxic volatile organic compounds like polychlorinated dibenzodioxins and dibenzofurans into the lower troposphere (Camilleri and Vella, 2010). These compounds have a high residence time in the atmosphere and travel long distances from the place of emission (Klima et al., 2020). The several constituents in the firecrackers on burning lead to an increment in particulate matter < 2.5 micrometer (PM_{2.5}) levels (Liu et al., 2019). The perturbations caused by the firework explosions enhance the particulate matter < 10 micrometer

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(PM₁₀) and black carbon levels in the atmosphere (Majumdar and Gavane, 2017). The aerosol concentrations and sulfur and nitrogen oxides (SO₂ and NO₂) exhibit a sharp increase after the burning of firecrackers (Chhabra et al., 2020). Attri et al. (2001) emphasized that the explosion of firecrackers produces ozone gas (O₃) even in the absence of nitrogen oxides. Other resultant gases like carbon monoxide (CO), carbon dioxide (CO₂), and ammonia (NH₃) could significantly pollute the atmosphere due to these fireworks (Sawani et al., 2019). Several pieces of research indicated the human health risks associated with the celebration using firecrackers (Garaga and Kota, 2018; 2020; Prabhu et al., 2019; Gowda et al., 2020; Sahu et al., 2020). High levels of particulate matter (both PM₁₀ and PM_{2.5}) in the ambient air cause an array of respiratory problems and cardiovascular disorders (Yunesian et al., 2019). Different types of asthma and obstructive pulmonary diseases are associated with pollutants like SO₂ and NO₂ (Greenberg et al., 2017). Moreover, many of these air pollutants are active carcinogens (Gadi et al., 2019; Kulshreshtha et al., 2021).

At present, the novel coronavirus (COVID-19) is the biggest threat to human society across the globe (Khot and Nadkar, 2020; Rahimi and Abadi, 2020; Zheng, 2020). As of May 2021, the outbreak of this virus led to almost 3.5 million official casualties (<https://www.worldometers.info/coronavirus>), and many remain uncounted (Wang et al., 2021). The spread of various strains of this virus has worsened the situation globally (Bappy et al., 2021; Duggan et al., 2021; Sarkar et al., 2021). The associated invasive fungal coinfections like Black, White, and Yellow fungus have become another point of severe concern while combatting this pandemic (Moorthy et al., 2021; Nori et al., 2021). Recent studies suggest that air pollution aggravates COVID-19 infectivity and mortality (Coccia, 2021). Travaglio et al. (2021) established a strong link between PM_{2.5} and the spread of coronavirus. Recent research suggests that NO₂ potentially enhances the fatality rate due to COVID-19 infections (Ogen 2020), and O₃ acts as a carrier and an incubator for coronaviruses (Zoran et al., 2020). India is one of the worst-affected countries struggling to cope with this pandemic (Kumari et al., 2021). With an intent to disrupt the spread of COVID-19 viruses, the Government of India announced a nationwide lockdown on 24 March 2020. Initially, the lockdown was for three weeks (Phase 1); however, the Government of India extended the lockdown for three more phases till 31 May 2020 (Soni, 2021). Keeping in mind the economic requirements, from 1 June 2020, some relaxations came into the scenario, and through several unlock phases, the society started resuming the pre-COVID-19 life (Gangwar and Ray, 2021). Several studies indicated that the COVID-19-induced nationwide lockdown drastically improved the air quality in all corners of the country (Bera et al., 2021; Mandal et al., 2021; Naqvi et al., 2021; Ravindra et al., 2021; Sathe et al., 2021). The complete stop of vehicular locomotion and a halt to all sorts of industrial activities reduced the pollutant levels below the thresholds in many Indian cities during the lockdown phases. In the year 2020, Diwali was on 14 November. During this time of 2020, India was slowly but steadily recovering from the first wave of COVID-19 spread. Keeping in view the adversities that enhanced air pollution can cause, the Honorable Supreme Court made an embargo on the sale or use of all kinds of firecrackers from midnight of 9 November 2020 to 30 November 2020 in all cities and towns across the country. Despite many such initiatives, the nation witnessed an unpredicted second wave from March 2021 (Mallapaty, 2021), reaching a peak in May 2021 when infection cases were 0.4 million people per day. However, previous studies reported a 34% decline in pollutants during Diwali due to a similar environment-friendly ban imposed on the free sale of firecrackers in the year 2018 (Kulshreshtha et al., 2021). Concerning this background, the present study aimed to analyze and compare the air quality of three megacities of India (Mumbai, Delhi, and Kolkata) on the auspicious day of Diwali in the year 2020 (in the presence of the COVID-19 pandemic) and the previous year 2019 (in the absence of pandemic). These three megacities are the most populous and most polluted in the entire country. The scientific literature indicates that the burning of

firecrackers in these densely populated regions significantly deteriorates the ambient air quality (Chatterjee et al., 2013; Anand et al., 2019; Sawani et al., 2019; Chattopadhyay and Shaw, 2021). We hypothesized that amidst an ongoing pandemic, people in large numbers must have restrained from burning firecrackers during Diwali and led to a lower degree of enhancement in pollutant levels than the observations in the previous year when there was no pandemic in the nation. Several studies focused on the effect of burning firecrackers on the air quality during Diwali. Similarly, the air pollution levels under a pandemic-induced lockdown have also received substantial attention from the scientific community in the recent past. However, none of these studies considered the effects of Diwali celebrations amidst a pandemic scenario. This study reports for the first time the degree of air pollutant levels from three megacities of India during Diwali amidst the COVID-19 pandemic. The findings of this study are expected to guide the policy managers of such populous and polluted cities to look for avenues to create a sustainable urban atmospheric environment. Since air quality management in urban setups has become a challenge for present-day urban planners, studies like this can act as eye-openers for both stakeholders and policymakers.

2. Material and methods

2.1. Study area

Mumbai, Delhi, and Kolkata are the three most populous megacities in west, north, and east India, respectively (Fig. 1). Mumbai, the largest megacity (also known as the business capital of India) located on the west coast (adjacent to the Arabian Sea) with an elevation of 14 m above the mean sea level (MSL) (Sansare and Mhaske, 2020), hosts a population of around 21 million (Brackowski et al., 2018). Delhi (the capital of India), situated in the Gangetic plains of northern India, has an elevation of 225 m and shelters a population of 16.8 million (Bhattacharyya and Prasad, 2020). Kolkata (also known as the cultural capital of India), situated in the eastern part of the country at 9 m above MSL, shelters a population of 14.1 million (Sikarwar, 2020). Mumbai and Kolkata experience a typical tropical climate throughout the annual cycle, whereas Delhi's climatic nature varied between humid subtropical and semi-arid. India experiences three seasons, namely the pre-monsoon (February to May), monsoon (June to September), and post-monsoon (October to January). Diwali dates around October or November are typical post-monsoon months with comparatively dry conditions but not as chilling as the winter months (December and January). Air temperature in October and November varies from ~20 °C to ~26 °C in Delhi, ~25 °C to ~28 °C in Kolkata, and ~28 °C to ~29 °C in Mumbai (IMD, 2015). The mean relative humidity in October and November ranges from 44% to 48%, 67% to 75%, and 65% to 72% in Delhi, Kolkata, and Mumbai, respectively (IMD, 2015). Similarly, mean rainfall ranges between 4.9 mm and 14.3 mm (Delhi), 35 mm and 180 mm (Kolkata), and 15 mm and 78 mm (Mumbai) (IMD, 2015). In October and November, Mumbai and Kolkata experience warm days, and the nights are soothing. However, in Delhi, the temperature tends to drop in November, leading to quite chilling nights. The atmospheric stability during these two months is relatively higher in Delhi than in Mumbai and Kolkata (Dutta et al., 2021; Sonwani et al., 2021).

2.2. Data used

This study considered seven parameters, namely PM_{2.5}, PM₁₀, NO₂, NH₃, SO₂, CO, and O₃. The ground-monitored data of these pollutants were downloaded from the central pollution control board (CPCB) official website (https://app.cpcbcr.com/AQI_India/). Delhi, Kolkata, and Mumbai have 36, 10, and 10 automatic air monitoring stations (AAMS), respectively. We downloaded the data for all of these stations. Diwali was on 27 October and 14 November in 2019 and 2020, respectively. Usually, people burn the firecrackers from the evening and

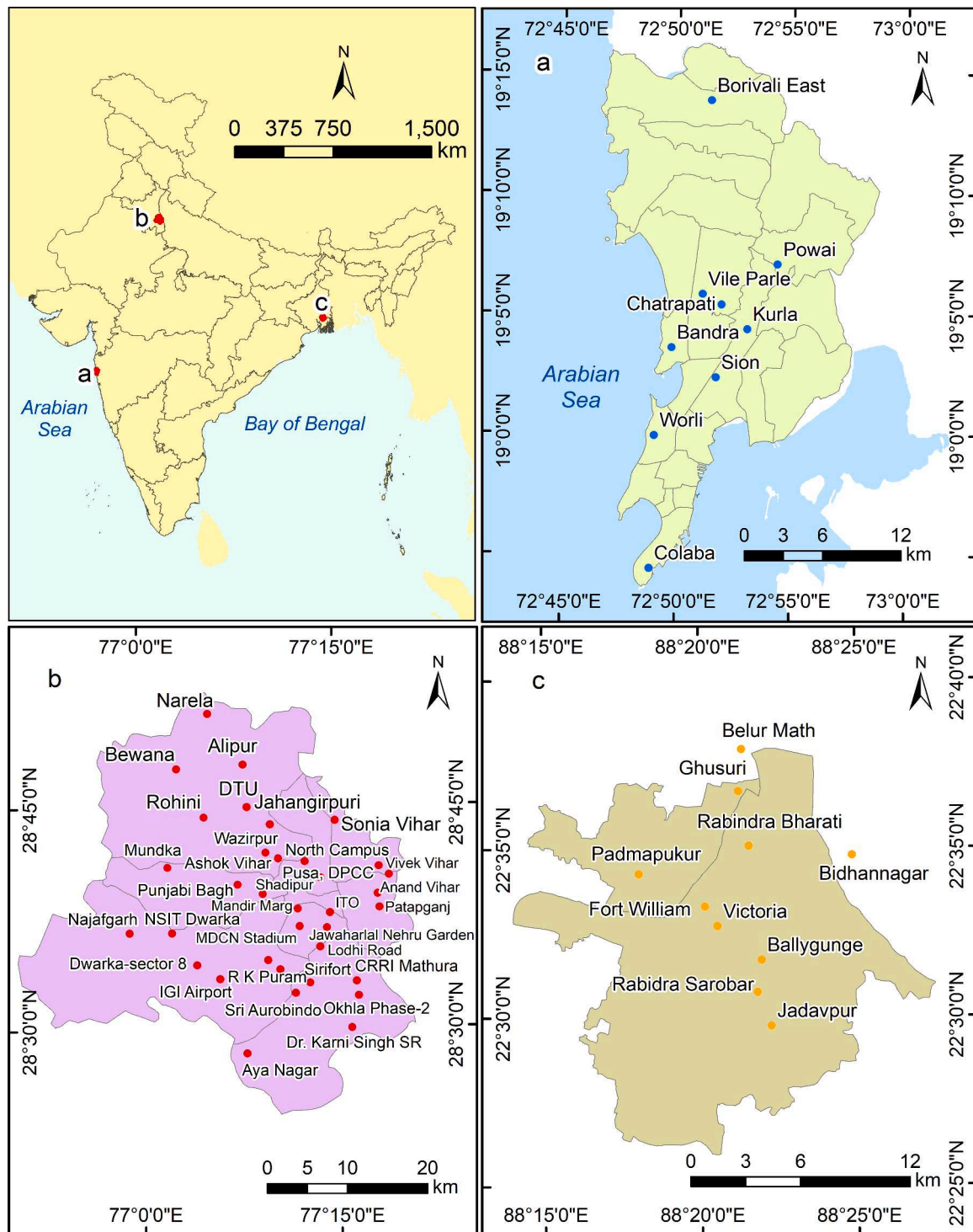


Fig. 1. The study area map showing the locations of automatic air monitoring stations (AAMS) of (a) Mumbai, (b) Delhi, and (c) Kolkata.

continues till midnight. All these AAMSs keep a record of hourly data. We extracted the data on all the seven pollutant concentrations at midnight (IST 12:00 am) on both of the dates. We considered 24 h average data (01:00 am to 12:00 am) for each Diwali date. To understand the rate of increase in pollutant concentrations and its short-term residence time in the atmosphere, we extracted and analyzed the pre-Diwali and post-Diwali data on the pollutant concentration for both years. The 24 h averaged data for the dates 20, 24, 25, and 26 October 2019, and 07, 11, 12, 13 November 2020 were analyzed to characterize the pre-Diwali pollution scenario. Similarly, the 24 h averaged data for

the dates 28, 29, 30 October, and 03 November 2019, and 15, 16, 17, and 21 November 2020 were analyzed to understand the post-Diwali pollution levels. The logic behind selecting these pre-Diwali and post-Diwali dates was to analyze and compare the pollutant levels just three days ago and just after three days after the event. In addition, we considered a date which was seven days before and seven days after the Diwali date to examine the residual effect in the atmosphere. Daily meteorological data of five parameters, namely ambient air temperature, relative humidity, wind velocity, wind direction, and precipitation for the same dates were retrieved from the weather stations deployed at

the Chhatrapati Shivaji International Airport Station (18.97°N, 72.83°E) for Mumbai; Safdarjung Airport Station (28.59°N, 77.21°E) for Delhi, and Mohestola, Behala Airport Station (22.54°N, 88.3 °E) for Kolkata (Weather Underground portal; <https://www.wunderground.com/>).

2.3. Methodology

The net change in each of the AAMS was computed to analyze the changes in pollutants concentrations between the different dates within the respective years. The composite mean of the net changes observed in all the AAMS was further computed for each megacity. The calculation of pre-Diwali to Diwali changes in pollutant concentrations followed two ways: (i) the difference between pollutant concentration in the Diwali date and seven days before Diwali and (ii) the difference between pollutant concentration in the Diwali date and the average pollutant concentration of three days before Diwali. Similarly, the Diwali to post-Diwali changes were also computed by analyzing (i) the difference between pollutant concentration in the Diwali date and seven days after Diwali and (ii) the difference between pollutant levels in the Diwali date and the average pollutant concentration of three days after Diwali. The box plots were prepared to represent the difference in ambient air pollutant concentration observed between Diwali 2019 and Diwali 2020 for each of the parameters. The error bars show the minimum and maximum observed in the respective dates. The intersection of the boxes denotes the median. The lower and the upper end of the boxes signify the first quartile and third quartile, respectively.

To characterize and compare the overall pollution scenario at the different selected dates, we computed the air quality index (AQI) according to the formula detailed in CPCB (2014). AQI formulation mainly includes two steps: (i) formation of sub-indices (for each pollutant) and (ii) identifying the worst sub-index that serves as AQI. We computed the sub-indices of seven pollutants at each station based on the 24 h average data (only CO and O₃ were 8 h average) and health break-point range. AQI computation needs PM_{2.5}, PM₁₀, NO₂, NH₃, SO₂, CO, O₃, and Pb as input parameters, of which at least three pollutant concentrations should be available and must include either PM_{2.5} or PM₁₀. The AQI has six categories: good, satisfactory, moderate, poor, very poor, and severe, based on a scale of 0 to 500 (Table 1). However, 24 h average data was not available for all the AAMS. Data from two AAMS (Kurla and Sion) for Mumbai, ten AAMS (AnandVihar, Ashok Vihar, Bewana, DTU, Jawaharlal Nehru Garden, Mundka, Patapganj, Pusa, Rohini, and Sonia Vihar) for Delhi and three AAMS for Kolkata (Ballygunge, Fort William, and Ghusuri) were utilized to compute the AQI for the selected dates.

2.3.1. Spatial interpolation of data

We analyzed the spatial variation in the air pollutants and AQI within each of the megacities using the ESRI ArcGIS 10.5 platform. The interpolation technique based on the Inverse Distance Weighting (IDW) interpolator and linear combination model was applied (Mandal et al.,

2021). IDW is a deterministic approach to compute the values at un-measured locations based on the measured values at distinct locations (Xie et al., 2017). The interpolation algorithm depends on a probabilistic logic that the influence of the values at a nearer location is higher than that of a distant location to an unknown location (Jumaah et al., 2019). In this method, the weights assigned to an unknown location are a function of the distance from the other locations having measured data (Vorapracha et al. 2015). The inverse of the squared distance between the unknown and the known locations forms the weight matrix, based on which the interpolation is carried out (Lloyd 2010). Jha et al. (2011) inferred that the IDW technique is best-suited for the Indian air pollution scenario and leads to fewer error margins than other conventional interpolation approaches.

2.3.2. Deweathering of air-pollutant data

Due to limited meteorological data and unavailability of meteorological data for all the AAMS, we deweathered one station's pollutant data from each of the three megacities, namely Anand Vihar (Delhi), Chatrapati (Mumbai), and Rabindra Sarobar (Kolkata). We used a generalized additive model to normalize the air pollutant data, i.e., to remove the effect of meteorological variables from the air pollutant concentrations (Munir et al., 2021). This model is a user-friendly and explainable machine-learning tool that removes the hindrances posed by non-linear variations in meteorological conditions to infer the time series variability of air pollutant concentrations (Ropkins and Tate, 2021; Solberg et al., 2021). The flexible approach of this model has made it quite popular among researchers (Dehghan et al., 2018; Li et al., 2019; Ravindra et al. 2019a), as it smoothes the air pollutant data negating the effect of weather variability and seasonal change. The normalization or deweathering of air pollutant data was carried out by implementing the mixed gam computation vehicle (mgcv) data tool (Wood, 2020) in the R programming language (R Core Team, 2020) by taking into consideration six parameters, namely air temperature, relative humidity, wind velocity, wind direction, precipitation, and day of the year. Parameters like solar radiation and cloud cover were excluded due to the unavailability of data.

2.3.3. Statistical analyze

Shapiro-Wilk test enabled us to check the normality of the data. An independent samples Student's *t* test indicated the difference in mean air pollutant levels among the cities and the two different years. Pearson correlation coefficient facilitated us to examine the inter-relationship between the ambient air pollutants. All statistical analyses involved using the SPSS software (version 16, SPSS Inc.). The results were considered significant at $p < 0.05$.

Table 1

Major pollutants concentration, national AQI classes, national AQI category and health impacts. Source: CPCB (2014) and Mahato et al. (2020).

Concentration range*							AQI	AQI category	Health Impact
PM _{2.5} (µg/m ³)	PM ₁₀ (µg/m ³)	NO ₂ (µg/m ³)	NH ₃ (µg/m ³)	SO ₂ (µg/m ³)	CO (mg/m ³)	O ₃ (µg/m ³)			
0–30	0–50	0–40	0–200	0–40	0–1.0	0–50	0–50	Good	Minimal impact
31–60	51–100	41–80	201–400	41–80	1.1–2.0	51–100	51–100	Satisfactory	Minor breathing discomfort to sensitive people
61–90	101–250	81–180	401–800	81–380	2.1–10	101–168	101–200	Moderately polluted	Breathing discomfort to the people with lung disease
91–120	251–350	181–280	801–1200	381–800	10.1–17	169–208	201–300	Poor	Breathing discomfort to people with prolonged exposure
121–250	351–430	281–400	1201–1800	801–1600	17–34	209–748	301–400	Very poor	Respiratory illness to the people on prolonged exposure
> 250	> 430	> 400	> 1800	> 1600	> 34	> 748	401–500	Severe	Respiratory illness to the people on prolonged exposure

* 24 h values for PM_{2.5}, PM₁₀, NO₂, NH₃, SO₂, and 8 h values for CO and O₃

3. Results and discussion

3.1. Air pollutant concentrations in Diwali 2019 and 2020

To break the chain of the spread of COVID-19 viruses, the Government of India (GOI) announced nationwide lockdown, initially for 21 days, starting from 24 March 2020, and Diwali was on 14 November 2020, which was during the sixth unlock phase, after four successive periods of lockdown and five unlock periods. The average concentrations of PM₁₀ and CO during Diwali were above the respective permissible limits in all three cities in 2019 and 2020 (Table 2). Such high concentrations of PM₁₀ prevailed in other cities of India, like Varanasi (Kumar et al., 2020), Nagpur (Rao et al., 2012), and Guwahati (Garaga and Kota, 2018), when there was no pandemic. PM_{2.5} was also beyond the limit in 2019 and 2020. The only exception was Mumbai which exhibited below threshold concentration during the Diwali of 2019. Fireworks consist of several fuels, oxidants, agglutinants, propellants, and coloring agents, which enhance the PM load in the ground-level atmosphere on burning (Hoyos et al., 2020). Recent pieces of research strongly indicated that both PM₁₀ and PM_{2.5} facilitate the spread of coronavirus (Setti et al., 2020) and lead to increased casualties (Yao et al., 2020). Tropospheric O₃ was below the permissible limit in all the instances, except for Kolkata, which experienced a higher concentration on Diwali 2020. Similarly, NO₂ levels were within the threshold limits on all occasions, except for Delhi during Diwali 2020. Usually, O₃ production takes place during the daytime due to the photochemical reactions involving NO₂. However, Attri et al. (2001) argued that several fireworks produce sufficient light to facilitate nighttime photochemical oxidation of NO₂ to produce O₃. Preliminary research indicated a high possibility of enhanced spread of coronavirus under elevated NO₂ and O₃ concentrations in respirable air (Ogen, 2020; Zoran et al., 2020). NH₃ and SO₂ were below the permissible limit in all three cities in both of the years. These observations indicate that there was no significant reduction in air pollutant levels during Diwali amidst the pandemic. On the contrary, to our surprise, almost all the pollutants showed elevated concentrations in Diwali 2020 compared to Diwali 2019. The only exception was Kolkata, where the mean concentration of PM_{2.5} in 2020 was lower than that observed in 2019. The difference in PM_{2.5} between Diwali 2019 and Diwali 2020 was statistically significant in Delhi and Mumbai ($p < 0.05$). For PM₁₀, NO₂, CO, and O₃, this difference was significant in all three cities ($p < 0.01$). The increase in NH₃ concentrations during 2020 was significant only in the case of Mumbai ($p < 0.05$), whereas for SO₂, the increase was significant for Delhi and Kolkata ($p < 0.05$). Delhi witnessed the highest increment in PM_{2.5} (125.42 µg/m³), PM₁₀ (142.84 µg/m³), and CO (52.84 mg/m³), while Kolkata recorded the highest in O₃ (38.20 µg/m³) between the Diwali of 2019 and 2020. Enhanced O₃ concentration in Kolkata indicates that

firecrackers that produce light were burnt in plenty, which in turn, facilitated the production of O₃ during nighttime. Similar observations were reported in previous studies (Attri et al., 2001; Nishanth et al., 2012). However, some contradicting opinions suggest that the burning of firecrackers produces several volatile organic compounds like phenol and benzene that mimic the O₃ during nighttime (Xu et al., 2018). The percentage of increase (in 2020 compared to 2019) in CO was 47% in Delhi, followed by Mumbai (39%) and Kolkata (20%). Though O₃ did not cross the permissible limit in Mumbai and Delhi, the increase in its concentration during the Diwali of the pandemic year was 68% in Mumbai, followed by Kolkata (58%) and Delhi (8%). PM_{2.5} and PM₁₀ recorded a nearly 30% increase in Mumbai and Delhi. CPCB computes the AQI from all the pollutant concentrations, and it serves as a holistic air pollution indicator. The AQI magnitudes indicate that Delhi was the most polluted megacity, followed by Kolkata and Mumbai during the Diwali of both years. Fig. 2 shows the ambient air pollutants' minimum, first quartile, median, third quartile, and maximum values on the day of Diwali in 2019 and 2020. Fig. 2a shows that all these five statistical measures of PM_{2.5} increased in 2020 for Mumbai and Delhi, except for Kolkata. Fig. 2b–d show an overall increase of PM₁₀, NO₂, and NH₃ during Diwali 2020 in all three cities. Fig. 2e shows that the first and third quartiles of SO₂ were almost the same in Mumbai in 2019 and 2020. Kolkata and Delhi showed a similar scenario in the case of CO (Fig. 2f) and O₃ (Fig. 2g), respectively. Fig. 2h shows the descriptive statistics of AQI.

3.2. Changes in air pollutant concentrations between pre-Diwali and post-Diwali

Tables 3 and 4 show the changes in air pollutant concentrations during the pre-Diwali and post-Diwali phases in 2019 and 2020, respectively. PM_{2.5} increased substantially during Diwali 2019 from the pre-Diwali levels, and the concentrations remained elevated even three days after Diwali 2019. In Mumbai and Kolkata, the PM_{2.5} levels after seven days of Diwali almost went back to the pre-Diwali scenario. However, in Delhi, the concentration kept on increasing even after seven days from Diwali (Table 3). Earlier studies showed Delhi suffers from elevated PM_{2.5} concentrations, which hampers the air quality and visibility at the ground level (Kumar et al., 2007; Sahu and Kota, 2017). In the year 2020, an opposite trend prevailed. The PM_{2.5} concentrations increased in the post-Diwali phases in Mumbai and Kolkata and decreased in Delhi after seven days from Diwali (Table 4). PM₁₀, on the whole, significantly increased during Diwali in both the years than their pre-Diwali levels, and the concentrations remained high even after one week from Diwali. Delhi was the only exception in 2020, where the one-week later concentrations reached the pre-Diwali levels. NO₂ concentrations were higher during the three days after Diwali 2019 than

Table 2

Average concentration of air pollutants in Diwali of previous year (2019) and lockdown year (2020) across three megacities of India. Source: CPCB (2019; 2020)

Pollutants	Mumbai				Delhi				Kolkata				Permissible Limit*
	27 Oct, 2019	14 Nov, 2020	Net Variation	%	27 Oct, 2019	14 Nov, 2020	Net Variation	%	27 Oct, 2019	14 Nov, 2020	Net Variation	%	
PM _{2.5} (µg/m ³)	72.88	106.13	33.25	31.33	317.00	442.89	125.89	28.42	138.70	128.20	-10.50	-8.19	60
PM ₁₀ (µg/m ³)	72.00	108.80	36.80	33.82	265.97	408.81	142.84	34.94	105.50	128.50	23.00	17.90	100
NO ₂ (µg/m ³)	20.00	41.86	21.86	52.22	62.03	91.68	29.65	32.34	33.60	45.10	11.50	25.50	80
NH ₃ (µg/m ³)	5.25	10.69	5.44	50.90	10.07	13.74	3.67	26.71	3.10	5.20	2.10	40.38	400
SO ₂ (µg/m ³)	11.67	13.92	2.25	16.17	15.03	21.59	6.56	30.38	14.56	22.20	7.64	34.43	80
CO (mg/m ³)	28.00	45.57	17.57	38.56	58.69	111.53	52.84	47.38	33.60	42.00	8.40	20.00	2
O ₃ (µg/m ³)	15.44	48.07	32.63	67.87	54.38	59.11	4.73	8.00	27.90	66.10	38.20	57.79	60
AQI	81.13	122.77			314.66	455.61			138.70	139.40			

* 24 h values for PM_{2.5}, PM₁₀, NO₂, NH₃, SO₂, and 8 h values for CO and O₃

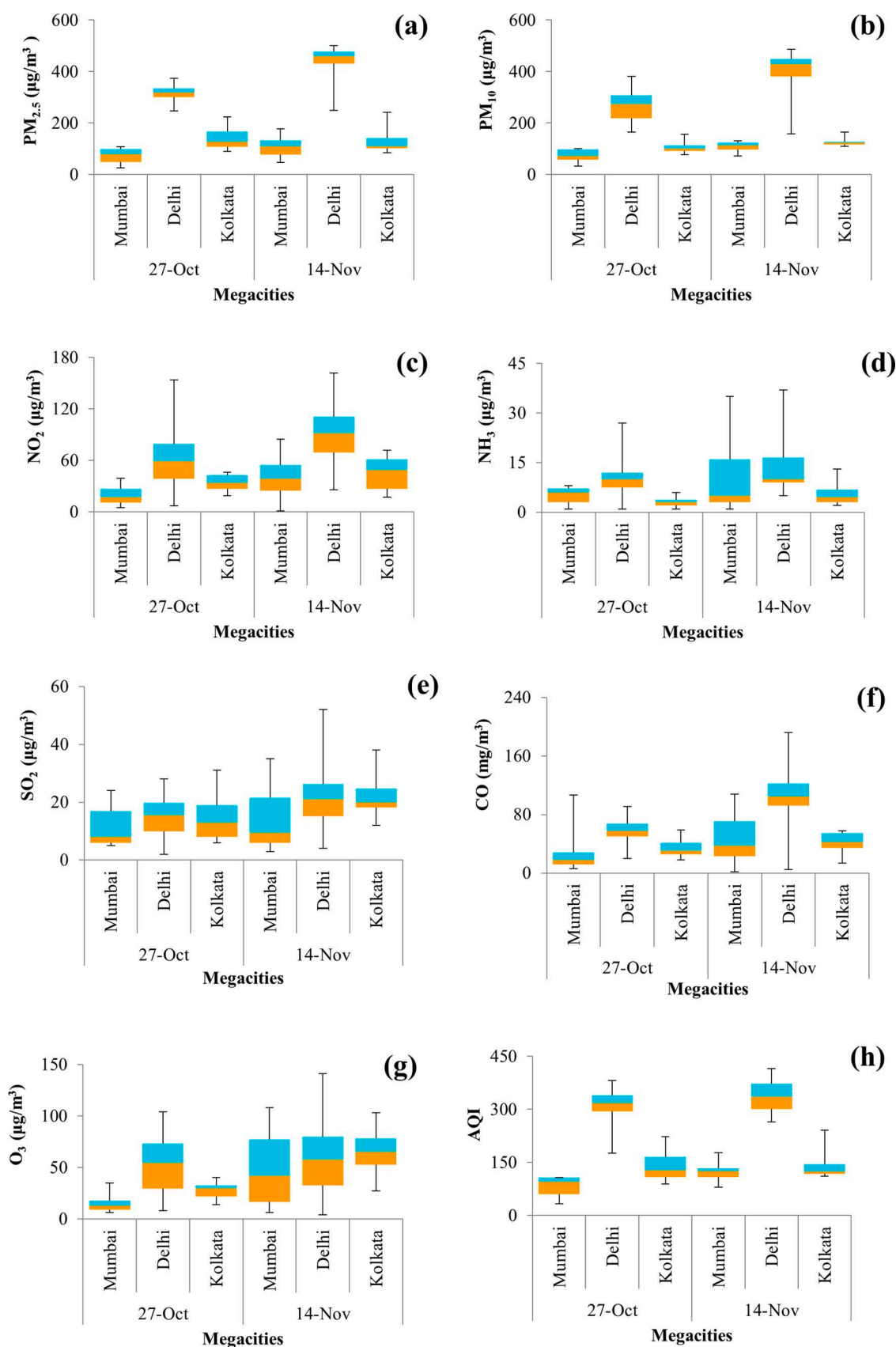


Fig. 2. The box plot showing the concentrations of (a) PM_{2.5}, (b) PM₁₀, (c) NO₂, (d) NH₃, (e) SO₂, (f) CO, (g) O₃, and (h) AQI at Mumbai, Delhi, and Kolkata on the day of Diwali in two consecutive years (27 Oct of 2019 and 14 Nov 2020). The error bars show the maximum and minimum values. The boxes show the first quartile, median, and third quartile from bottom to top.

Table 3

Average concentration of ambient air pollutants in a normal year (2019) of three megacities of India, on the Day of Diwali, seven days before and after Diwali, and three-day average before and after Diwali. Source: [CPCB \(2021\)](#)

Pollutants	Megacity	Normal year, 2019					Variation							
		20-Oct	Avg. of 24-26 Oct	27 Oct, 2019	Avg. of 28-30 Oct	03-Nov	27 Oct and 20 Oct		27 Oct and Avg. of 24-26 Oct		Avg. of 28-30 Oct and 27 Oct		3 Nov and 27 Oct	
							Net	%	Net	%	Net	%	Net	%
PM _{2.5} (µg/m ³)	Mumbai	29.00	48.67	102.00	144.17	39.00	73.00	71.57	53.33	52.29	42.17	29.25	-63.00	-43.70
	Delhi	227.40	316.50	329.90	418.83	461.20	102.50	30.95	13.40	3.92	88.93	21.38	131.30	27.18
	Kolkata	203.00	77.67	176.00	226.89	186.33	-27.00	-15.34	98.33	55.87	50.89	22.43	10.33	4.55
PM ₁₀ (µg/m ³)	Mumbai	37.00	69.67	98.00	121.00	85.00	61.00	62.24	28.34	28.91	23.00	19.01	-13.00	-10.74
	Delhi	181.40	280.03	296.00	414.87	435.11	114.60	37.81	15.97	6.10	118.87	26.38	141.33	30.97
	Kolkata	135.67	75.56	121.00	170.33	159.67	-14.67	-12.12	45.44	37.56	49.33	28.96	38.67	22.70
NO ₂ (µg/m ³)	Mumbai	44.50	46.17	36.00	46.25	34.50	-8.50	-23.61	-10.17	-28.25	10.25	22.16	-1.50	-3.24
	Delhi	62.60	74.90	68.70	77.65	69.90	6.10	10.27	-6.20	-8.28	8.95	11.03	1.20	3.89
	Kolkata	39.00	30.00	33.67	35.78	42.33	-5.33	-15.84	3.67	10.89	2.11	5.90	8.67	24.22
NH ₃ (µg/m ³)	Mumbai	6.00	3.67	8.00	5.34	7.00	2.00	25.00	4.34	54.19	-2.67	-49.95	-1.00	-18.74
	Delhi	8.90	10.50	11.70	14.20	15.20	2.80	24.71	1.20	10.49	2.50	14.83	3.50	18.58
	Kolkata	3.67	4.22	4.33	5.33	6.33	0.67	15.38	0.11	2.56	1.00	18.75	2.00	37.50
SO ₂ (µg/m ³)	Mumbai	16.50	18.00	19.00	17.34	17.50	2.50	13.16	1.00	5.25	-1.67	-9.60	-1.50	-8.65
	Delhi	13.89	16.83	15.40	16.23	11.56	1.22	4.53	-1.43	-8.53	0.83	8.00	-4.22	-23.59
	Kolkata	9.00	8.33	14.33	18.89	22.67	5.33	37.21	6.00	41.86	4.56	24.12	8.33	44.12
CO (mg/m ³)	Mumbai	32.00	30.33	12.00	17.67	21.00	-20.00	-166.67	-18.33	-152.78	5.67	32.07	9.00	50.95
	Delhi	52.00	72.27	61.80	108.23	89.60	9.80	14.14	-10.47	-11.66	46.43	40.83	27.80	23.13
	Kolkata	39.67	29.78	35.33	35.11	32.33	-4.33	-12.26	5.56	15.73	-0.22	-0.64	-3.00	-8.54
O ₃ (µg/m ³)	Mumbai	10.00	10.83	20.00	42.42	14.50	10.00	50.00	9.17	45.84	22.42	52.85	-5.50	-12.97
	Delhi	42.30	69.03	52.90	34.33	21.70	10.60	18.56	-16.13	-25.25	-18.57	-35.49	-31.20	-50.65
	Kolkata	48.00	23.44	28.00	39.22	41.33	-20.00	-71.43	4.56	16.27	11.22	28.61	13.33	33.99
AQI	Mumbai	44.50	71.50	102.00	144.17	85.00								
	Delhi	227.40	312.43	331.30	417.27	479.80								
	Kolkata	203.00	87.67	176.00	226.89	186.33								

Table 4

Average concentration of ambient air pollutants for COVID-19 year (2020) of three megacities of India, on the Day of Diwali, seven days before and after Diwali, and three-day average before and after Diwali. Source: [CPCB \(2021\)](#)

Pollutants	Megacity	COVID-19 year, 2020					Variation							
		07-Nov	Avg. of 11-13 Nov	14-Nov	Avg. of 15-17 Nov	21-Nov	14 Nov and 7 Nov		14 Nov and Avg. of 11-13 Nov		Avg. of 15-17 Nov and 14 Nov		21 Nov and 14 Nov	
							Net	%	Net	%	Net	%	Net	%
PM _{2.5} (µg/m ³)	Mumbai	128.00	169.17	120.50	170.50	197.50	-7.50	-5.29	-48.67	-39.85	50.00	28.49	77.00	39.05
	Delhi	432.00	348.93	473.00	258.93	265.90	41.00	8.67	135.33	28.61	-235.67	-49.82	-245.00	-51.80
	Kolkata	247.00	103.89	130.33	222.45	226.67	-116.67	-89.51	26.44	20.29	92.11	41.41	96.33	42.50
PM ₁₀ (µg/m ³)	Mumbai	168.50	168.84	127.00	152.50	190.50	-41.50	-32.52	-41.84	-33.13	25.50	16.51	63.50	33.27
	Delhi	410.60	329.00	437.10	183.92	202.60	26.50	5.93	108.10	25.13	-253.18	-57.92	-234.50	-53.65
	Kolkata	139.33	121.33	138.33	182.00	164.33	-1.00	-0.72	17.00	12.29	43.67	23.99	26.00	15.82
NO ₂ (µg/m ³)	Mumbai	82.50	67.00	58.50	55.67	82.50	-24.00	-46.79	-8.50	-15.44	-2.84	-2.49	24.00	29.78
	Delhi	83.60	80.07	91.90	54.60	56.90	8.30	7.42	11.83	10.57	-37.30	-40.59	-35.00	-38.08
	Kolkata	48.33	55.00	50.00	67.55	50.33	1.67	-2.18	-5.00	-9.71	17.55	24.98	0.33	0.53
NH ₃ (µg/m ³)	Mumbai	33.00	36.33	18.50	17.17	13.50	-14.50	-36.88	-17.83	-41.66	-1.33	-2.05	-5.00	-32.07
	Delhi	12.00	12.90	12.80	11.65	7.90	0.80	5.38	-0.10	-2.11	-1.91	-7.49	-4.90	-31.19
	Kolkata	3.33	9.44	5.00	5.11	5.67	1.67	33.33	-4.44	-88.89	0.11	2.22	0.67	11.76
SO ₂ (µg/m ³)	Mumbai	8.50	9.50	9.50	9.83	12.00	1.00	11.67	0.00	0.00	0.33	3.38	2.50	20.63
	Delhi	20.40	19.53	20.90	19.33	14.80	0.50	1.04	1.37	5.33	-1.57	-8.44	-6.10	-30.48
	Kolkata	32.67	23.44	26.00	36.22	30.00	-6.67	-25.64	2.56	9.83	10.22	28.22	4.00	13.33
CO (mg/m ³)	Mumbai	44.00	39.84	26.50	35.50	43.50	-17.50	-73.61	-13.34	-51.59	9.00	25.90	17.00	38.95
	Delhi	90.10	104.33	114.00	55.03	60.30	23.90	19.52	9.67	7.36	-58.97	-88.46	-53.70	-45.33
	Kolkata	56.33	36.78	43.00	46.00	79.67	-13.33	-31.01	6.22	14.47	3.00	6.52	36.67	46.03
O ₃ (µg/m ³)	Mumbai	33.00	67.33	74.50	74.50	62.00	41.50	52.93	7.17	7.56	0.00	0.12	-12.50	-21.37
	Delhi	45.20	53.47	72.30	43.10	43.90	27.10	38.02	27.00	24.70	-43.67	-36.77	-43.00	-37.63
	Kolkata	78.33	48.56	70.67	59.55	18.33	-7.67	-10.85	22.11	31.29	-11.11	-18.66	-52.33	-285.45
AQI	Mumbai	168.50	186.67	128.00	178.67	197.50								
	Delhi	429.30	351.33	472.40	262.73	265.90								
	Kolkata	247.00	118.11	139.67	223.44	226.67								

those observed on the day of Diwali. The increment in motor vehicles already leads to enhanced NO₂ levels in Delhi, and the firecrackers add to that pre-existing load ([Ganguly, 2009](#); [Singh et al., 2010](#)). However, in the year 2020, Delhi was the only city to have higher NO₂ levels on the day of Diwali compared to both pre-Diwali and post-Diwali days. Mumbai and Kolkata did not record any significant variation across the pre-Diwali to post-Diwali phases. NH₃ and SO₂ concentrations showed a consistently increasing trend from pre-Diwali to Diwali to post-Diwali transitions of 2019 and 2020 in Delhi and Kolkata; however, Mumbai

showed a gradual decrease after Diwali. CO showed a steady decline in concentration from pre-Diwali to post-Diwali in Mumbai and Kolkata; however, Delhi showed a consistent increase during both years. O₃ increased in Delhi and Mumbai during the same timeframe; however, Kolkata recorded a steady decrease. AQI, which indicates the overall air pollution scenario showed a consistent increase in Delhi even after seven days of Diwali in 2020; however, in the other two cities, the AQI gradually decreased after the event. Among the changes in air pollutant concentrations observed between different phases, the one between the

pre-three-day average and the day of Diwali acts as a proxy of the degree of fireworks. This difference was substantially high for all the pollutants in Delhi during the Diwali of 2020, except for NH_3 . These results suggest that during the sixth unlock phase, vehicular and industrial emissions were substantially high. These emissions enhanced the pollutant loads in the lower atmosphere even before Diwali (Maji et al., 2021; Ravindra et al., 2021). During Diwali 2020, the general mass believed that India has already seen the worst and is steadily overcoming the pandemic. This perception might have led the people to celebrate Diwali like in earlier years. Please see supplementary files (Table S1-S6) for detailed station-wise results.

3.3. Spatial distribution of AQI

Figs. 3 and 4 show the spatial distribution of all the pollutants in Delhi during the pre-Diwali, Diwali, and post-Diwali days of 2019 and 2020. Visual inspection of these maps portrays that the northern and eastern parts exhibited more pollutant concentrations than the rest of the city. This observation could be due to the high population density in these areas. Tyagi et al. (2016) and Garg et al. (2019) observed similar spatial variability of several air pollutants in Delhi. These maps also show that Delhi has a higher residence time of pollutants. Padmanabhamurthy and Ravichandran (1990) and Guttikunda and Gurjar (2012) observed the less dispersal capacity of pollutants in Delhi, especially during the post-monsoon season. Almost all the parameters showed higher color bars in the post-Diwali phase. These figures also exhibit a significant increase in pollutant levels during the pre-Diwali to Diwali transition in both years. This observation indicates no noticeable change in the burning of firecrackers during the pandemic-stricken Diwali of 2020. There was a marked difference in the transition phases of 2019 and 2020 in Kolkata (Figs. 5 and 6) and Mumbai (Figs. 7 and 8). Many of the pollutants, except for PM_{10} , showed a lower color bar in the Diwali of 2020 than the pre-Diwali scenario. This observation illustrates that the restricted burning of firecrackers took place to some

extent in Kolkata and Mumbai. The northern and eastern parts of Kolkata showed higher pollution owing to the high population density. The western part of Mumbai exhibited lesser air pollution, which could be due to adjacent sea breezes that help in the quick dilution of the pollutants.

3.4. Correlation between air-pollutants

Taking into account the pollutant data for both the years, PM_{10} and $\text{PM}_{2.5}$ showed a significant positive correlation ($p < 0.01$) in all three megacities (Fig. 9). Zhou et al. (2016) observed a similar correlation all through China. They observed that these two different-sized particulate matters have almost similar compositions. O_3 and NO_2 showed a significant positive correlation with particulate matter levels in Mumbai. NO_2 is the principal catalyst in tropospheric O_3 production. Hence, usually, these two should exhibit a negative or no correlation at all. However, reports of positive and negative correlations between these pollutants exist in Indian cities (Venkitaswamy and Bhaskar, 2015). NO_2 and NH_3 showed similar correlation with PM_{10} and $\text{PM}_{2.5}$ ($p < 0.05$), respectively. However, no such correlation existed with O_3 . In Kolkata, SO_2 and NO_2 showed a significant positive correlation ($p < 0.05$). These two gases can severely deteriorate lung activity and affect the bronchioles (Moseholm et al., 1993), which lead to complexities in COVID-19 infected patients.

3.5. Role of meteorological parameter and interrelationship between pollutants

Table 5 shows the average, minimum, and maximum air temperature, relative humidity, wind speed, and total rainfall for the three megacities during Diwali 2019 and 2020. There was no statistically significant difference in the mean air temperature of Mumbai and Kolkata between the Diwali of two consecutive years. However, Delhi recorded a significantly lower air temperature in 2020 than in 2019.

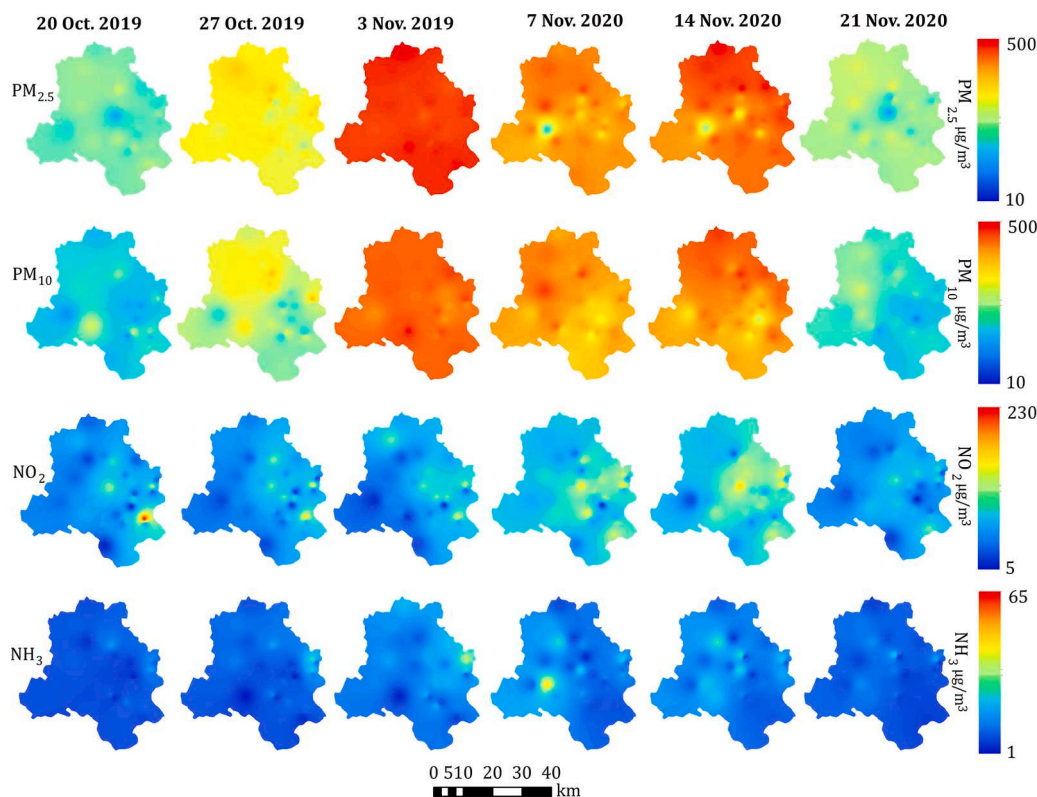


Fig. 3. The spatial distribution of $\text{PM}_{2.5}$, PM_{10} , NO_2 , and NH_3 in Delhi on the seventh day before Diwali, on Diwali, and the seventh day after Diwali of 2019 and 2020.

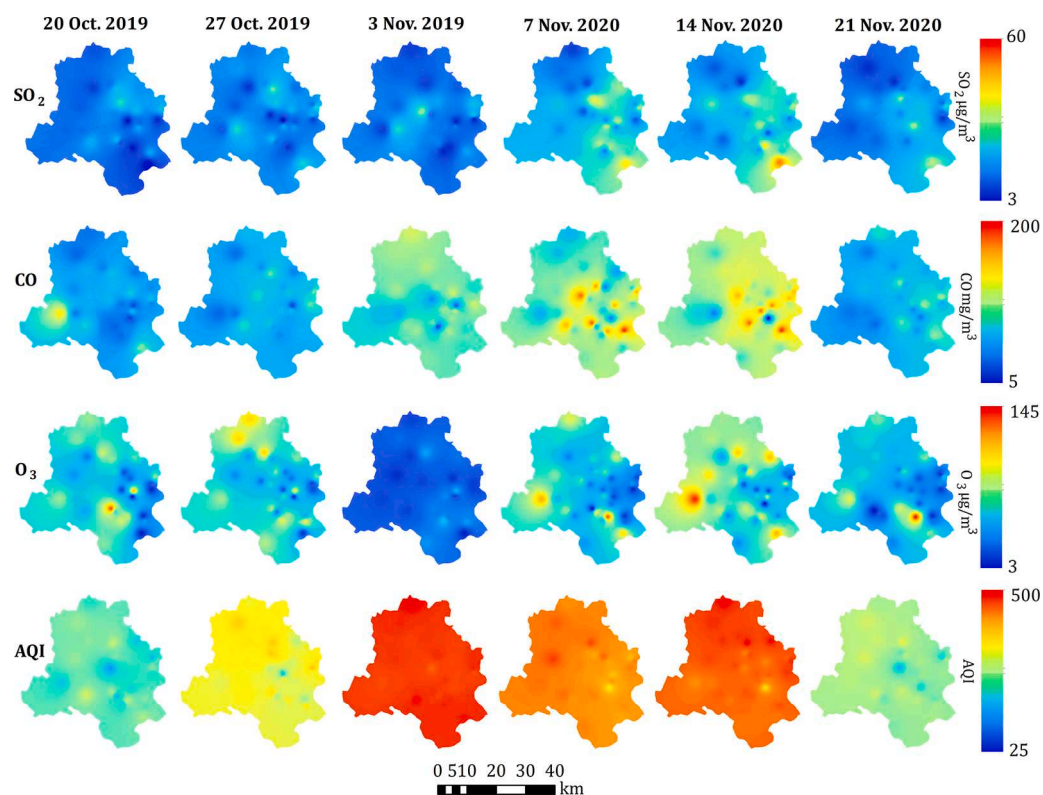


Fig. 4. The spatial distribution of SO_2 , CO , O_3 , and AQI in Delhi on the seventh day before Diwali, on Diwali, and the seventh day after Diwali of 2019 and 2020.

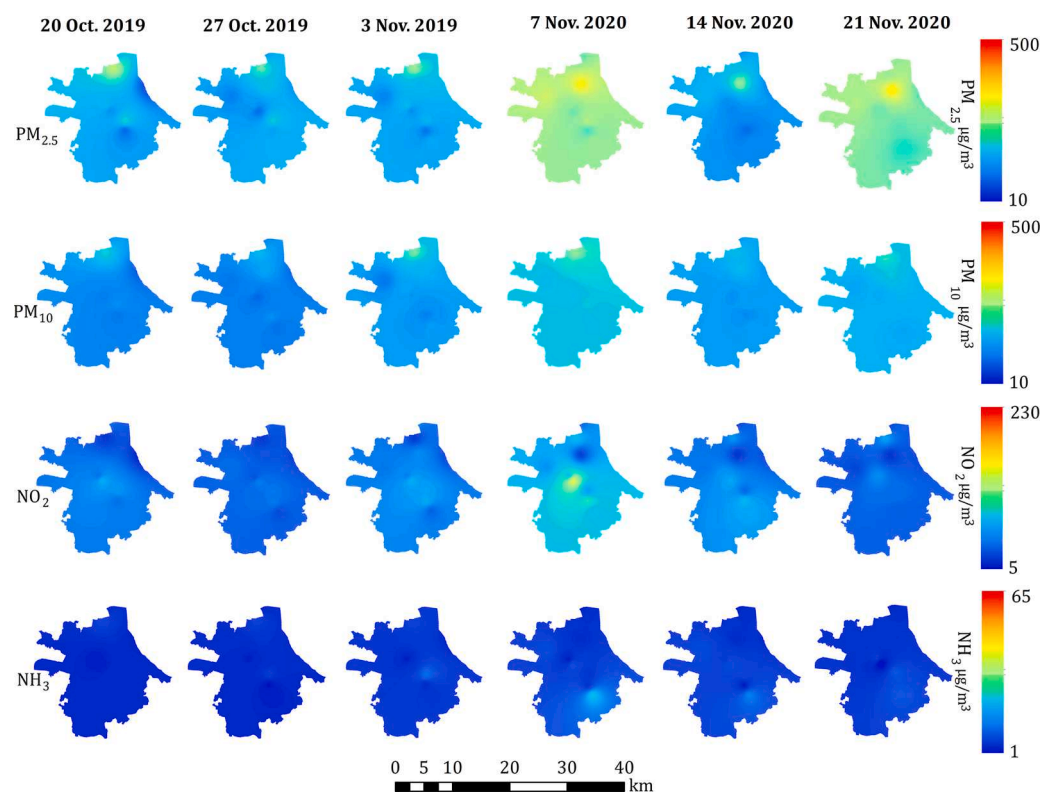


Fig. 5. The spatial distribution of $\text{PM}_{2.5}$, PM_{10} , NO_2 , and NH_3 in Kolkata on the seventh day before Diwali, on Diwali, and the seventh day after Diwali of 2019 and 2020.

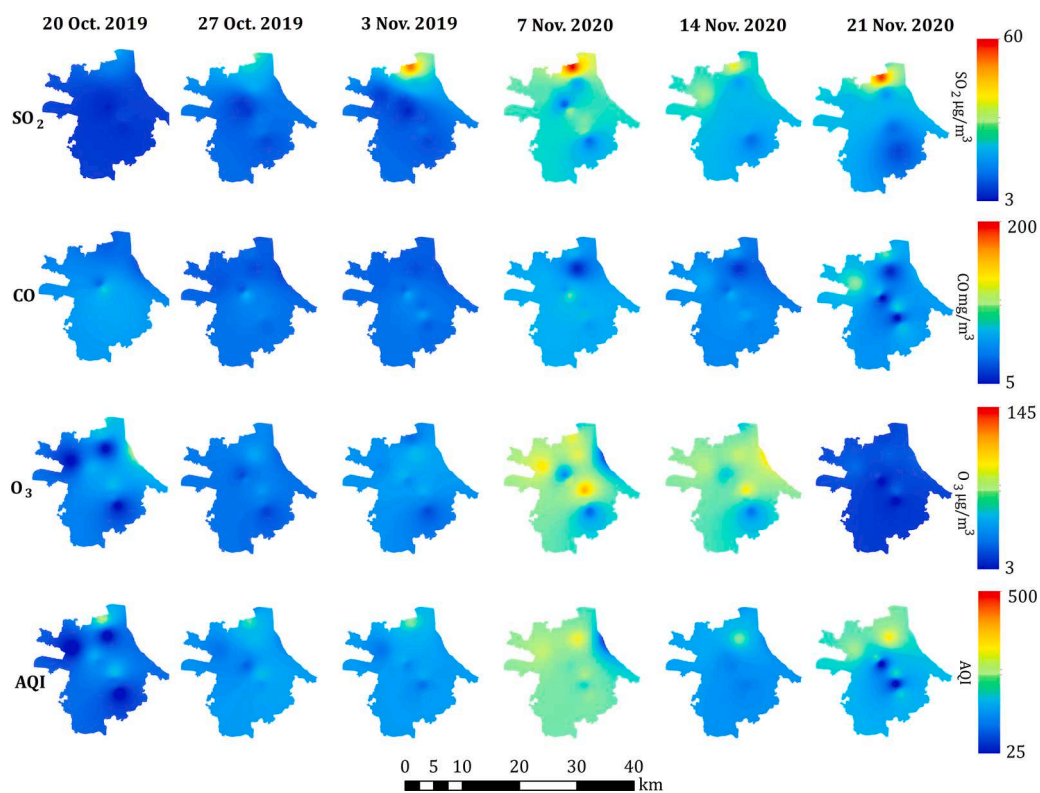


Fig. 6. The spatial distribution of SO₂, CO, O₃, and AQI in Kolkata on the seventh day before Diwali, on Diwali, and the seventh day after Diwali of 2019 and 2020.

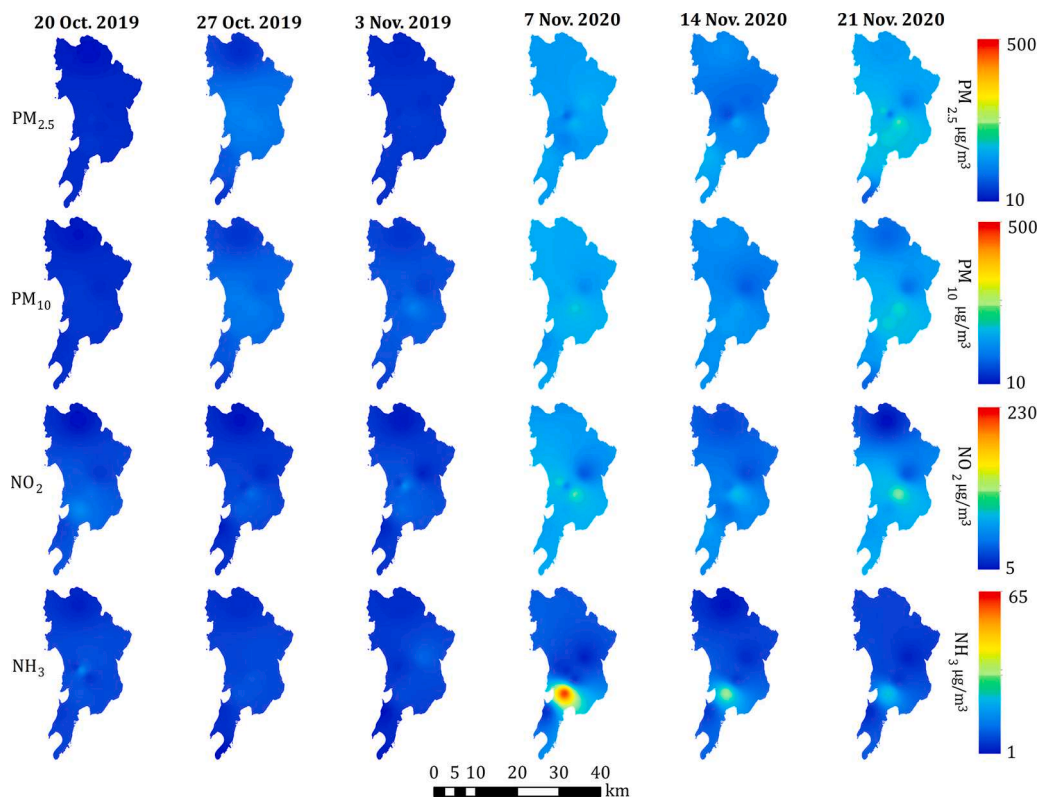


Fig. 7. The spatial distribution of PM_{2.5}, PM₁₀, NO₂, and NH₃ in Mumbai on the seventh day before Diwali, on Diwali, and the seventh day after Diwali of 2019 and 2020.

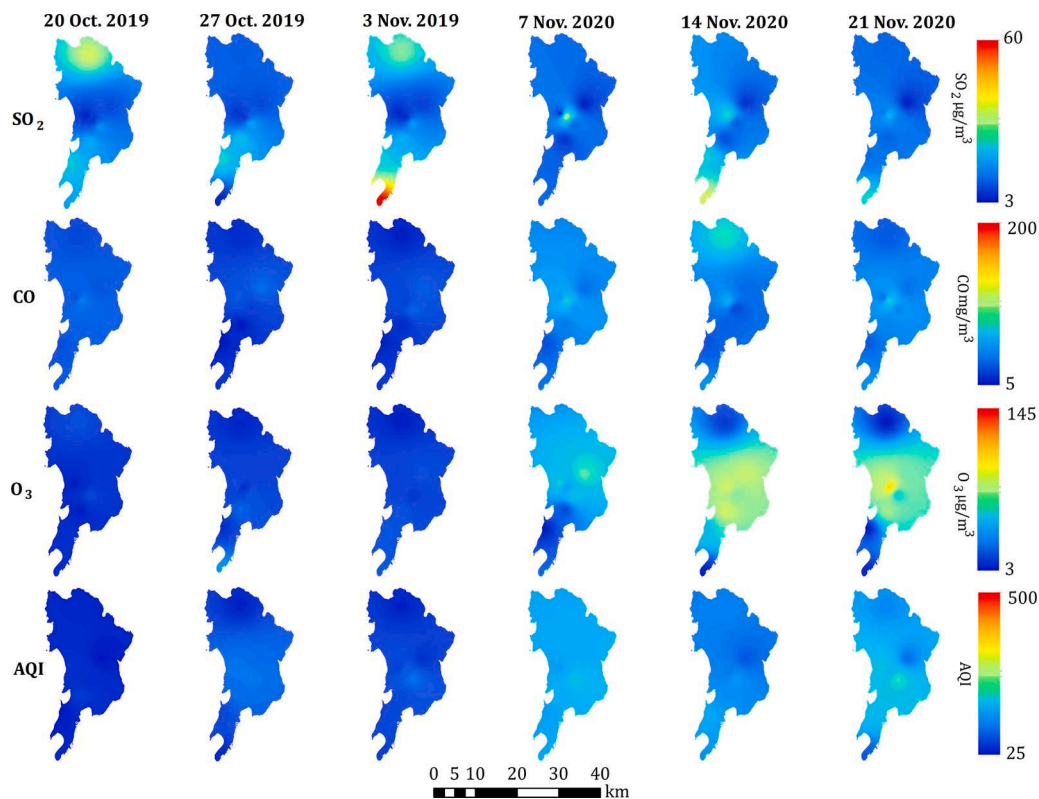


Fig. 8. The spatial distribution of SO_2 , CO , O_3 , and AQI in Mumbai on the seventh day before Diwali, on Diwali, and the seventh day after Diwali of 2019 and 2020.

Tiwary et al. (2015; 2018) observed that low temperature facilitated higher pollutant load in Delhi, which could be one of the reasons behind higher concentration in the pre-Diwali phase of 2020 than in 2019. Increased air temperature leads to an unstable atmosphere, which allows the pollutants to dilute quickly (Cichowicz et al., 2017; Ravindra et al., 2019b). Relative humidity was comparatively lower in Mumbai and Kolkata (during Diwali) in the year 2020 compared to that in 2019. Lower air moisture content facilitates the enhancement of pollutants and vice-versa (Lou et al., 2017; Munir et al., 2017). Delhi recorded an elevated relative humidity during Diwali in the pandemic year. This increased humidity should have significantly reduced the pollutant load. However, a pre-Diwali to Diwali increment in pollutant concentrations, despite having high moisture, indicates the rampant burning of firecrackers in Delhi. Wind speed showed no significant difference between the 2019 and 2020 Diwali phases in Mumbai and Kolkata. Usually, higher wind speeds dissipate the pollutant concentrations, and a reverse scenario indicates a stable atmosphere, which retains the pollutants (Li et al., 2020). In Delhi, the wind speed was much less in the 2020 Diwali than in the 2019 Diwali. This low wind speed could have worsened the scenario in Diwali 2020. The wind direction exhibited significant changes between pre-Diwali and Diwali dates in few instances (see Figs. S1 and S2 in the supplementary files). In the year 2019, the predominant northeastern winds in pre-Diwali changed to the northwest on the Day of Diwali in Mumbai. The wind from the western end came over the sea, thus, helped to dilute the air pollutant concentrations. In the pandemic year 2020, no such wind reversal was observed in Mumbai. Delhi witnessed a change in wind direction from the south-southeast to the north-northwest during the pre-Diwali to Diwali. The wind that flows from the north comes from the Himalayas and is usually cold that led to a drop in temperature. This reduction in temperature might have facilitated enhanced air pollutant levels in Delhi during Diwali. There was no rainfall in any of the cities in the 2019 and 2020 Diwali phases. Precipitation usually dampens the pollutant loads (Yoo et al., 2014); however, it could not play any role in regulating the pollutant load in

any of the two years. Thus, analyzing the observations altogether, we could partially accept our hypothesis for Mumbai and Kolkata. However, the results from Delhi outrightly rejected the same we framed for this study.

3.6. Interpretation of changes in deweathered air pollutant data

The deweathered data from one station each for both the years 2019 and 2020 is listed in Table 6. Due to the unavailability of complete meteorological dataset, we had to restrict ourselves to only one station. However, the deweathered data also portrayed similar trend to that observed in the case of unprocessed data set for all the stations. Deweathered $\text{PM}_{2.5}$ and PM_{10} showed significant increase from the pre-Diwali (seventh day before Diwali) to Diwali in Delhi in the pandemic year 2020. Mumbai and Kolkata also exhibited an increase in these two parameters. However, the magnitude of increase was much less compared to that of Delhi. NO_2 did not show any significant increase between the pre-Diwali and Diwali in any of the megacities during 2020, but NH_3 exhibited an increase in Mumbai and Delhi. SO_2 increased substantially from pre-Diwali to Diwali in Delhi and Kolkata during the pandemic year 2020. CO did not show any significant variation. However, SO_2 and O_3 increased substantially on the Diwali day of 2020 in Delhi. O_3 levels in particular exhibited an increase in all the three cities during Diwali in the pandemic year. Only one station's data is not sufficient to infer for the entire city. However, the deweathered data also indicates that Delhi is the worst affected among the three megacities considered in this study.

4. Conclusion

The present study compared the air pollution levels (during Diwali) of three megacities of India, namely Delhi, Mumbai, and Kolkata, between a non-pandemic year (2019) and a pandemic-stricken year (2020). Previous studies indicated that burning firecrackers leads to

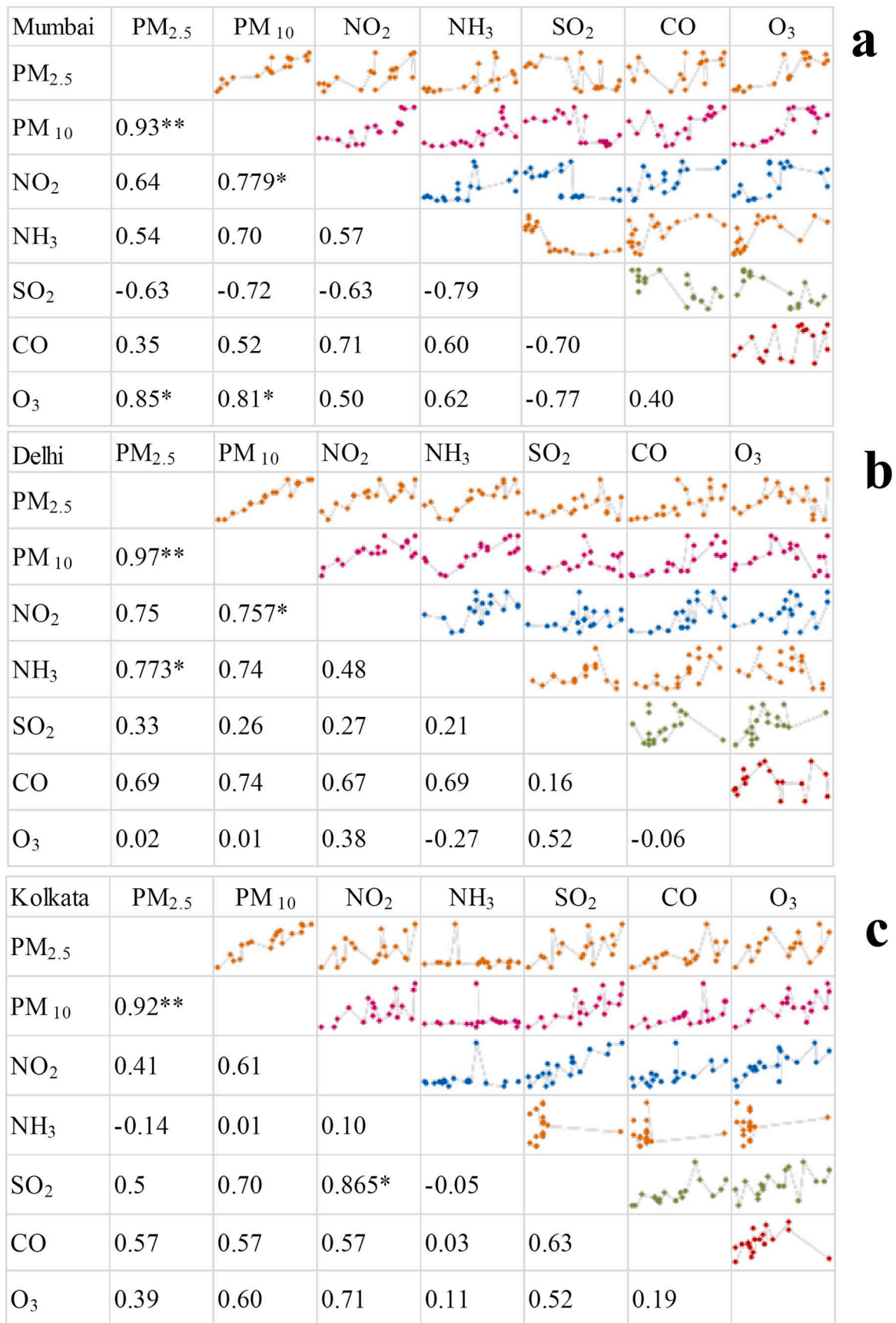


Fig. 9. The correlation matrices and scatter plots between the seven air pollutants across the megacities of (a) Mumbai, (b) Delhi, and (c) Kolkata [*Correlation is significant at the 0.05 level (2-tailed); **Correlation is significant at the 0.01 level (2-tailed)].

Table 5

Meteorological parameters (air temperature, relative humidity, and wind speed) and total rainfall observed during Diwali 2019 and 2020 in Mumbai, Delhi, and Kolkata.

Megacity	Date	Temperature (° F)			Humidity (%)			Wind Speed (mph)			Rainfall (inch) Total
		Max	Avg	Min	Max	Avg	Min	Max	Avg	Min	
Mumbai	27 Oct 2019	90	84.2	77	89	70.6	52	13	6.1	0	0
	14 Nov 2020	95	86.3	79	74	58.5	41	13	6	0	0
Delhi	27 Oct 2019	89	74.9	65	90	66.4	34	9	3.3	0	0
	14 Nov 2020	82	63.9	54	97	82.1	53	2	0.6	0	0
Kolkata	27 Oct 2019	84	76.6	70	94	86.7	66	9	4.4	0	0
	14 Nov 2020	88	77.3	66	83	61.5	43	10	4.2	0	0

Table 6

Deweathered air pollutant data of Delhi (Anand Vihar), Mumbai (Chhatrapati), and Kolkata (Rabindra Sarobar) of the seventh day before Diwali, on Diwali, and the seventh day after Diwali in the year 2019 and 2020.

	PM _{2.5}	PM ₁₀	NO ₂	NH ₃	SO ₂	CO	O ₃
Delhi							
20-10-2019 (pre-Diwali)	147.3	312.5	105.2	68.4	7.0	32.0	21.5
27-10-2019 (Diwali)	282.7	479.7	115.4	94.9	15.8	32.6	12.6
03-11-2019 (post-Diwali)	425.6	575.1	114.2	103.8	15.6	33.0	8.5
07-11-2020 (pre-Diwali)	288.7	525.8	139.1	54.3	13.1	54.0	27.4
14-11-2020 (Diwali)	363.0	608.4	121.5	57.1	19.7	54.0	31.9
21-11-2020 (post-Diwali)	131.5	251.6	83.9	25.4	15.0	52.3	31.5
Mumbai							
20-10-2019 (pre-Diwali)	22.5	37.2	19.4	47.2	5.7	22.9	7.8
27-10-2019 (Diwali)	39.0	98.3	20.8	24.1	5.2	22.8	9.4
03-11-2019 (post-Diwali)	34.0	98.3	59.3	21.7	4.7	22.7	13.6
07-11-2020 (pre-Diwali)	36.9	153.9	39.5	14.7	29.2	25.7	45.0
14-11-2020 (Diwali)	43.8	160.1	39.3	21.8	20.8	25.6	61.8
21-11-2020 (post-Diwali)	52.3	148.1	38.5	26.2	16.2	26.6	51.3
Kolkata							
20-10-2019 (pre-Diwali)	51.6	97.0	37.7	12.4	4.7	31.7	36.6
27-10-2019 (Diwali)	93.7	130.4	41.2	7.4	9.4	32.8	27.2
03-11-2019 (post-Diwali)	75.5	150.2	61.6	10.0	11.3	33.0	34.3
07-11-2020 (pre-Diwali)	82.6	182.0	84.3	16.3	20.9	32.5	41.1
14-11-2020 (Diwali)	51.0	115.1	60.9	10.2	20.2	32.9	48.5
21-11-2020 (post-Diwali)	75.6	137.2	64.9	15.5	13.2	32.1	21.0

intense air pollution and that air pollution aids in the transmission of COVID-19 viruses. These generalized observations prompted us to examine the difference in air pollution levels during the Diwali of 2020 amidst the COVID-19 pandemic and compare the scenario with that of the previous year's Diwali when there was no pandemic. The study revealed elevated PM_{2.5}, PM₁₀, NO₂, NH₃, SO₂, CO, and O₃ concentrations in Diwali 2020 in all three cities, with very few exceptions. This observation primarily indicates that people celebrated Diwali with more vigor even under a pandemic-stricken condition. But, analyzing the pre-Diwali to Diwali changes in air pollutant concentrations for both years portrayed a lesser increase for most of the pollutants in Mumbai and Kolkata. Delhi was an exception, where the enhancement in pollutant levels during Diwali in 2020 was much higher than that observed in 2019. Earlier studies showed that Delhi is the worst sufferer of air pollution throughout the year. The present study indicated that despite air pollution problems, the people of Delhi did not restrict themselves from burning firecrackers even in a pandemic situation, which can significantly aggravate the pandemic and lead to more casualties. Moreover, Diwali in the pandemic year took place at such a time when the sixth unlock phase was in action. The nation was steadily going back to the pre-pandemic norms. The terror of coronavirus was slowly

dissipating at this point, which might have triggered the people to celebrate Diwali like before. In addition, an absence of awareness regarding the causal effects of burning firecrackers to promote the spread of the virus might have played a crucial role. In 2020, Diwali was in mid-November when air temperature and wind speed were lower than the Diwali held in October 2019. This difference in meteorological conditions might have led to a partial enhancement in the overall pollutant load in Delhi's ground-level atmosphere. However, the pre-Diwali to Diwali changes in pollutant concentrations indicated a dramatic increase, which could be possible only because of burning firecrackers. Mumbai and Kolkata did not record such magnitudes of increase in pollutant concentrations from pre-Diwali to Diwali. It is more concerning that Delhi retained the pollutant loads in the atmosphere even after one week of Diwali in 2020. The stable atmosphere of Delhi enhances the residence time of air pollutants, especially in the winter months, which comes just after Diwali. The PM₁₀, PM_{2.5}, and CO levels were above the permissible limit in all three megacities during the Diwali of 2019 and 2020. Besides Delhi, such elevated concentrations are a point of concern for Mumbai and Kolkata. However, the concentrations were almost double in Delhi compared to those observed in Mumbai and Kolkata. Recent studies indicated a sharp decline in air pollution levels during the nationwide lockdown. Diwali in 2020 was concomitant with the sixth unlock phase when the country was almost back on track recovering from the pandemic. In the next few months, India witnessed a deadly second wave of COVID-19 infections. Though Diwali is a one-day event, burning firecrackers for even one day can aggravate the infection scenario and deteriorate the ground-level atmosphere. Thus, we require stringent law enforcement against fireworks. It is undeniable that Diwali involves the religious sentiments of hundreds of millions of people across the country, and India has a long tradition of celebrating this festival in a harmonious mood. Any forceful prohibition might hurt the religious sentiments of many. Thus, the government should launch and promote non-polluting crackers at the earliest to save the environment from further deterioration. Besides Diwali, several other events involve burning firecrackers. Such initiatives can help improve the ambient air quality in the coming days to secure a sustainable atmospheric environment in these populous urban setups. The highly populous megacities require special attention in this regard. A control in the growth of population density and indiscriminate urban sprawl are badly needed to check the further deterioration of the lower atmosphere in these megacities of India.

Declaration of Competing Interest

The authors declare that they have no known competing conflict of interest.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jad.2018.04.077](https://doi.org/10.1016/j.jad.2018.04.077).

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